

DELHIVERY

NETWORK OPERATIONS

STRATEGY MEMO

Bottleneck Hub Identification & Corridor Specific Intervention Recommendations

TO Head of Network Operations	FROM Data Science Team	DATE June 2026	REF Graph Network Intelligence v2
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EXECUTIVE SUMMARY

This memo presents findings from a **graph based network analysis** of Delhivery's logistics operations across **1,600+ hubs and 15,695 trip segments**. Using directed graph centrality metrics and observed SLA data, we isolate five structural chokepoints and their highest impact corridors : the specific source to destination lanes driving the most delay through each hub. These corridors are the primary lever for intervention.

The network's delay problem is not uniform: **29.7% of all trips breach SLA**, yet the majority of delay hours concentrate on fewer than 30 corridors. Corridor level targeting rather than hub level blanket upgrades yields a higher return per rupee invested.

29.7% Overall SLA Breach Rate	1,073 hrs Total Network Delay Hours	685 Chronic Corridors (>1.2× OSRM)	5 of 1,600+ Hubs Flagged Critical
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HOW WE IDENTIFIED THESE HUBS

Each hub received a **Bottleneck Score (0–1)** combining two equal weight dimensions, both normalised to [0,1] before weighting:

STRUCTURAL RISK 50% of Bottleneck Score Graph topology metrics: betweenness centrality (50%), degree (25%), closeness (15%), inverse clustering (10%). A hub with high betweenness sits on many shortest paths : delay there cascades broadly.	PERFORMANCE RISK 50% of Bottleneck Score Operational metrics: total delay hours accumulated (40%) and that hub's % contribution to total network delay pool (60%). A hub must fail on both counts to surface here.
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TOP 5 BOTTLENECK HUBS

#	Hub	State	Score	Delay Hrs	Btw. Cent.	SLA	Classification
1	Gurgaon Bilaspur HB	Haryana	0.847	98.7	0.216	100%	Critical
2	Bangalore Nelmangala H	Karnataka	0.598	26.8	0.122	100%	Critical
3	Bhiwandi Mankoli HB	Maharashtra	0.532	60.4	0.056	100%	Critical
4	Kanpur Central H	Uttar Pradesh	0.415	34.5	0.033	100%	Critical
5	Ahmedabad East H	Gujarat	0.365	10.6	0.031	100%	Critical

HUB-BY-HUB ASSESSMENT & CORRIDOR-SPECIFIC INTERVENTIONS

1

Gurgaon Bilaspur HB | Haryana

Bottleneck Score: 0.847 — CRITICAL

The network's highest risk hub by a significant margin. Betweenness centrality of 0.216 : the highest of all flagged hubs - means Gurgaon Bilaspur sits on the shortest path for a disproportionate share of all origin destination pairs. At 98.7 accumulated delay hours and a 100% SLA breach rate, every trip routing through this hub is arriving late. The corridor Hub 14 to Hub 30 alone shows a 77.6× average delay ratio across 61 shipments, the worst-performing high volume corridor in the dataset.

Corridor (Source → Dest)	Delay Ratio	Volume	Recommended Intervention
Hub 14 → 30 (NH-48 approach)	77.6×	61 ships	Redirect 30 - 40% FTL volume to Manesar hub via NH-48 alternate; impose peak hour dispatch ban 08:00 - 11:00
Hub 30 → 14 (return corridor)	10.7×	58 ships	Convert carting to FTL dispatch on this lane; add dedicated unloading dock for return freight
Hub 30 → 623 (outbound spoke)	2.8×	2 ships	Monitor; reroute via Hub 94 if delay persists beyond next 4 week window

2

Bangalore Nelmangala H | Karnataka

Bottleneck Score: 0.598 — CRITICAL

Structural gateway for South India routes (betweenness 0.122). Failure mode is facility dwell-time, not corridor congestion — Hub 1059→4 shows an 8.9× delay ratio, pointing to one specific inbound lane as the root cause.

Corridor (Source → Dest)	Delay Ratio	Volume	Recommended Intervention
Hub 1059 → 4 (inbound critical)	8.9×	7 ships	Reroute this lane via Tumkur bypass; classify as Priority FTL with 2-hr guaranteed unload slot
Hub 5 → 4 (high volume inbound)	1.4×	68 ships	Stagger inbound windows across 06:00–08:00 and 14:00–16:00; upgrade bay capacity on receiving dock
Hub 4 → 5 (outbound)	1.3×	92 ships	Highest-volume corridor at this hub; convert to dedicated FTL run with fixed dispatch at 07:00 daily

3

Bhiwandi Mankoli HB | MaharashtraBottleneck Score: **0.532 — CRITICAL**

Bhiwandi is a known congestion point and the data confirms it: second-highest absolute delay hours (60.4 hrs) with a 2.65% SLA breach contribution to the total network delay pool — the highest network-wide bleed of any hub. Lower betweenness (0.056) relative to its delay burden identifies this as a primarily operational failure: the volume is manageable by topology, but the hub is not processing it effectively. The Hub 872 → 2 corridor registers a 15.4× delay ratio, pointing to a specific inbound lane as the primary driver.

Corridor (Source → Dest)	Delay Ratio	Volume	Recommended Intervention
Hub 872 → 2 (inbound extreme)	15.4×	9 ships	Immediate reroute via Hub 600 alternate gate; flag for audit — 15× ratio indicates possible data/routing config error
Hub 58 → 2 (inbound moderate)	1.3×	7 ships	Stagger FTL dispatch windows; activate Bhiwandi alternate unloading gate for this lane
Hub 2 → 32 (outbound)	1.4×	6 ships	Shift to morning FTL dispatch (05:30) to avoid peak NH-8 congestion window 08:00–10:00

4

Kanpur Central H | Uttar PradeshBottleneck Score: **0.415 — CRITICAL**

Average delay ratio of 5.24× signals severe OSRM miscalibration for UP arterial routes, not random variance. Hub 341→92 (24.9×) is the priority corridor fix; OSRM recalibration for UP lanes is the parallel intervention.

Corridor (Source → Dest)	Delay Ratio	Volume	Recommended Intervention
Hub 341 → 92 (via Kanpur, critical)	24.9×	7 ships	Reroute to bypass Kanpur Central via Unnao satellite hub; this is the highest-priority single corridor fix in UP
Hub 27 → outbound (Kanpur Central)	1.6×	est. 40+	Move carting volume to Unnao satellite; recalibrate OSRM time estimates for UP-region corridors
Kanpur inbound from NCR	~5×	est. 30+	Commission OSRM corridor audit for UP lanes; implement buffer time adjustment (+40%) pending recalibration

5

Ahmedabad East H | GujaratBottleneck Score: **0.365 — CRITICAL**

Ahmedabad East rounds out the top five with 10.6 delay hours and a 100% breach rate. Its SLA contribution registers near-zero network-wide, making it the lowest-urgency of the five — but a 100% breach rate at a Gujarat gateway hub with established betweenness (0.031) will compound as west-India volume grows. The Hub 19 → 176 corridor (10.6× delay ratio) feeding into the Ahmedabad cluster is the single fix that will have outsized impact relative to investment.

Corridor (Source → Dest)	Delay Ratio	Volume	Recommended Intervention
Hub 19 → 176 (inbound extreme)	10.6×	1 ship	Isolated extreme — investigate routing config; likely OSRM miscalibration on this specific lane
Hub 199 → 176 (inbound moderate)	3.4×	1 ship	Flag for capacity audit; add secondary unloading bay to handle Gujarat inbound growth
Hub 710 → 176 (regular inbound)	2.9×	7 ships	Stagger inbound windows; schedule FTL arrivals before 07:00 to avoid Ahmedabad AM congestion

REVENUE AT RISK — TOP 3 HUB UPGRADE SCENARIO

Revenue at risk was quantified using a two-component model: **SLA penalty costs** (₹200/late shipment, industry standard for Indian B2B logistics contracts) and **customer churn cost** (2% churn probability per breach × ₹5,000 LTV proxy). A 60% breach-rate reduction post-upgrade is applied as the intervention benchmark (BCG Supply Chain, 2022).

Hub	SLA Breaches	Penalty Cost	Churn Cost	Total Revenue at Risk
Gurgaon Bilaspur HB	406	₹81,200	₹40,600	₹1,21,800
Bangalore Nelmangala H	319	₹63,800	₹31,900	₹95,700
Bhiwandi Mankoli HB	60	₹12,000	₹6,000	₹18,000
TOTAL (Top 3)	785	₹1,57,000	₹78,500	₹2,35,500

471 Breaches prevented (60% fix)	₹1,41,300 Revenue recovered	185+ hrs Delay hours eliminated
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RECOMMENDED INTERVENTIONS — PRIORITY ORDER

#	Hub	Corridor-Level Action	Timeline	Est. Delay Reduction
1	Gurgaon Bilaspur HB	Parallel route NH-48; FTL ban Hub 14→30 peak hours; Manesar diversion 30→40% volume	4–6 wks	35–45 hrs/month
2	Bhiwandi Mankoli HB	Stagger FTL windows; alternate gate Hub 872→2; morning dispatch Hub 2→32	2–4 wks	25–30 hrs/month
3	Kanpur Central H	Reroute Hub 341→92 via Unnao; OSRM recalibration for UP-region corridors	3–5 wks	18–22 hrs/month
4	Bangalore Nelmangala H	Tumkur bypass for Hub 1059→4; dedicated FTL slot Hub 4→5 at 07:00 daily	6–8 wks	15–20 hrs/month
5	Ahmedabad East H	Secondary unloading bay; stagger Hub 710→176 arrivals before 07:00	6–10 wks	5–8 hrs/month